SECTION I. PHYSICAL CHEMISTRY

Problem 1 (author Kandaskalov D.V.)

1. We know the approximate half-life of the nucleus, this allows us to calculate the possible values of the atomic mass of the nucleus, calculating the number of atoms decayed in $t = 2$ years ($N_0 - N$):

$$\ln\frac{N}{N_0} = -\lambda\cdot t = -\frac{\ln 2}{t_{1/2}}\,t \rightarrow N = N_0\exp\left(-\frac{\ln 2}{t_{1/2}}\,t\right) \rightarrow N_0 - N = N_0\left(1-\exp\left(-\frac{\ln 2}{t_{1/2}}\cdot t\right)\right).$$

Substitute the two extreme values (10 and 16 years) for the half-life:

$$N_0 - N = 2.82\cdot 10^{21}\left(1-\exp\left(-\frac{\ln 2}{10}\cdot 2\right)\right) = 0.365\cdot 10^{21}\ \text{nuclei},$$

$$N_0 - N = 2.82\cdot 10^{21}\left(1-\exp\left(-\frac{\ln 2}{16}\cdot 2\right)\right) = 0.234\cdot 10^{21}\ \text{nuclei}.$$

On the other hand, one can calculate the mass of decomposed nuclei **Y** (without knowing the meaning of the unit Ci). If the specific activity is 9650 Ci/g, then the loss of activity of 14.475 Ci should correspond to 14.475/9650 = 0.0015 g of the substance, from which the atomic mass **Y** can be estimated:

$$\frac{N_0 - N}{N_A} = \frac{\Delta m}{M} \rightarrow M = \frac{\Delta m\cdot N_A}{N_0 - N} = \frac{0.0015\cdot 6.02\cdot 10^{23}}{(0.234\text{ or }0.365)\cdot 10^{21}} = 2.47 - 3.85\ \text{g/mol}.$$

The only integer in the resulting interval is 3 g/mol (1 point), so we can find the exact half-life of the nucleus. First we write:

$$N_0 - N = \frac{\Delta m}{M}\cdot N_A = \frac{0.0015}{3}\cdot 6.02\cdot 10^{23} = 3.01\cdot 10^{20}\ \text{nuclei}$$

$$\rightarrow N = 2.82\cdot 10^{21} - 3.01\cdot 10^{20} = 2.52\cdot 10^{21}\ \text{nuclei} \rightarrow N/N_0 = 0.893.$$

Now calculate the exact half-life (1 point):

$$\ln\frac{N}{N_0} = -\frac{\ln 2}{t_{1/2}}\cdot t \rightarrow t_{1/2} = -\frac{\ln 2}{\ln N/N_0}\cdot t = -\frac{\ln 2}{\ln 0.893}\cdot 2 = 12.25\ \text{years}.$$

(2 points in total)

2. Let's calculate what part of the **Y** nuclei will decay in 2 months (0.167 years):

$$\ln\frac{N}{N_0} = -\lambda\cdot t = -\frac{\ln 2}{t_{1/2}}\,t \rightarrow N = N_0\exp\left(-\frac{\ln 2}{t_{1/2}}\,t\right) \rightarrow$$

$$\frac{N_0 - N}{N_0} = 1-\exp\left(-\frac{\ln 2}{t_{1/2}}\cdot t\right) = 1-\exp\left(-\frac{\ln 2}{12.25}\cdot 0{,}167\right) = 0.0041.$$

This part corresponds to 2.46 g and thus the loss of the budget will be $ 73,500. (1 point in total)

3. **a)** We know that the atomic mass of **Y** is 3, but we don't know the number of protons in the nucleus. Possible values: 1 (tritium), 2 (helium), or 3 (lithium). The last option can be excluded, since a nucleus consisting of only three protons does not exist. Let's take the reaction of artificial fusion of the **Y** nucleus: ${}^{2Z}_{Z}\mathbf{A} + {}^{1}_{0}n \rightarrow {}^{2Z'}_{Z'}\mathbf{Z} + {}^{3}_{Z''}\mathbf{Y}$. From here we get a system of equations:

$$\begin{cases} 2Z + 1 = 2Z' + 3 \\ Z + 0 = Z' + Z'' \end{cases} \Rightarrow \begin{cases} Z - Z' = 1 \\ Z - Z' = Z'' \end{cases} \Rightarrow Z'' = 1.$$

Thus, the nucleus **Y** - T (tritium) (1 point). Then the nucleus **X** must be hydrogen or deuterium. **b**) Write the reaction in the reactor: ${}^{3}_{1}\mathbf{T} + {}^{A}_{1}\mathbf{X} \rightarrow {}^{2Z'}_{Z'}\mathbf{Z} + {}^{1}_{0}n$. It turns out that $Z' = 2$, then it is helium and $A = 2$, that is, **X** is deuterium (**1 point**). Returning to the first reaction, we find that **A** is ^{6}Li. Then **X** - ^{2}D, **Y** - ^{3}T, **Z** - ^{4}He, **A** - ^{6}Li. We will get the following three reactions: (1 point for each reaction)

$${}^{6}_{3}\mathrm{Li} + {}^{1}_{0}n \rightarrow {}^{4}_{2}\mathrm{He} + {}^{3}_{1}\mathrm{T},\quad {}^{3}_{1}\mathrm{T} + {}^{2}_{1}\mathrm{D} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}n,\quad {}^{14}_{7}\mathrm{N} + {}^{1}_{0}n \rightarrow {}^{12}_{6}\mathrm{C} + {}^{3}_{1}\mathrm{T}.$$

(5 points in total)

4. Calculate the number of particles in the reactor through its volume: $V = 2\pi^2 R^2 r = 2\cdot\pi^2\cdot 9^2\cdot 0.52 = 831.4\ \text{m}^3$. Number of pairs of nuclei:

$$p = \frac{C\cdot V}{2} = \frac{10^{18}\cdot 831.4\cdot 10^3}{2} = 4.16\cdot 10^{21}\ \text{pairs}.$$

(1 point)

So the released energy

$$E_{tot} = p\cdot E_{p=1} = 4.15\cdot 10^{21}\cdot 3\cdot 10^{-12} = 1.25\cdot 10^{10}\ \text{J}.$$

(1 point)

Let's find the enthalpy of combustion of one mole of propane and one mole of butane to get the average value of the heat of combustion of 1 mole of a mixture (1: 1) of propane and butane: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$, $C_4H_{10} + 6.5O_2 \rightarrow 4CO_2 + 5H_2O$.

$$\Delta H_1 = 3\Delta_f H(CO_2) + 4\Delta_f H(H_2O) - \Delta_f H(C_3H_8) =$$

$$= 3\cdot(-393.5) + 4\cdot(-285.8) - (-103.9) = -2220\ \text{kJ/mol},$$

$$\Delta H_2 = 4\Delta_f H(CO_2) + 5\Delta_f H(H_2O) - \Delta_f H(C_4H_{10}) =$$

$$= 4\cdot(-393.5) + 5\cdot(-285.8) - (-126.2) = -2877\ \text{kJ/mol},$$

$$\Delta H_{mix} = 0.5(\Delta H_1 + \Delta H_2) = 0.5(-2220 - 2877) = -2548\ \text{kJ/mol},$$

$$n_{mix} = \frac{E_{tot}}{|\Delta H_{mix}|} = \frac{1.25\cdot 10^{10}}{|-2.55\cdot 10^6|} = 0.49\cdot 10^6\ \text{mol}.$$

(2 points in total)

The total mass of deuterium and tritium in the reactor is 34 mg and the energy effect of the reaction is equivalent to the combustion of 4900 moles of the mixture. It should be taken into account that the total energy costs for maintaining the plasma were not taken into account, which can be equivalent to 75–90% of the energy received.

5. Let us write the equation for the density of metal **M** and compound **N**:

$$\rho_M = \frac{M_M\cdot N_M}{V_M\cdot N_A},\qquad \rho_N = \frac{M_N\cdot N_N}{V_N\cdot N_A},\qquad \frac{\rho_M}{\rho_N} = \frac{M_M\cdot N_M\cdot V_N}{V_M\cdot M_N\cdot N_N} = 1.23.$$

(1 point)

We do not know which cubic cell the metal **M** has, then **NM** can be 1, 2 or 4 for cubic, body-centered and face-centered lattices. Let's leave it as a parameter. The formula unit of **N** is **MA**, where **M** is metal and **A** is the second element (we don't know if it's metal or non-metal), $N_N = 1$, and $M_N = M_M + x$. The unit cell volume of metal **M**, compounds **N** and their ratio will be:

$$V_M = a^3 = 3.16^3 = 31.55\ \text{Å}^3,$$

$$V_N = \frac{\sqrt{3}}{2}\cdot a^2\cdot c = \frac{\sqrt{3}}{2}\cdot 2.92^2\cdot 2.82 = 20.82\ \text{Å}^3,$$

$$\frac{V_N}{V_M} = = 0.660.$$

$$\frac{\rho_M}{\rho_N} = \frac{M_M\cdot N_M\cdot V_N}{V_M\cdot M_N\cdot N_N} = 0.660\cdot\frac{M_M}{M_M + x}\cdot\frac{N_M}{1} = 1.23,$$

$$\omega_M = \frac{M_M}{M_M + x} = \frac{1.239}{0.660\cdot N_M} = \frac{1.877}{N_M}.$$

(2 points for ω_M)

We have found that the mass fraction is in the range of 0.50-1.00. Since the value of N_M can only be equal to 2 (body-centered cell), the mass fraction of metal is 0.9386. That is, the atomic mass of **M** is 15.3 times greater than **A**. (1 point) Knowing that **M** can have an atomic mass of 140-260 (if we exclude radioactive elements, then not higher than 210), we conclude that the atomic mass of **A** lies in the range of 9.15-17 (not higher than 13.7 without radioactive elements). (1 point) So it could be B, C, (N, O, including radioactive elements). By simple selection, we get **N** - WC, **M** - W. (5 points in total)

Problem 2 (authors Rozantsev G.M., Shved E.N.)

1. As a result of hydrolysis of Me^{3+}, $Me^{3+}$and $MeOH^{2+}$ participate in the routes: $Me^{3+} + X^- \rightarrow MeX^{2+}$ (k_1) and $MeOH^{2+} + X^- \rightarrow MeOHX^+$ (k_2). (**1 point in total**)

2. Reaction rate of $[Fe(H_2O)_6]^{3+} + Cl^-$, proceeding in two routes, $r = k_1[Fe^{3+}]\,C(Cl^-) + k_2[FeOH^{2+}]\cdot C(Cl^-)$. Since $[Fe^{3+}]$ and $[FeOH^{2+}] >> [FeCl^{2+}]$ and $[FeOHCl^+]$, then the latter can be neglected and $C(Fe) = [Fe^{3+}] + [FeOH^{2+}] = [Fe^{3+}]([H^+] + K_a)/[H^+])$, $[Fe^{3+}] = [H^+]\cdot C(Fe)/([H^+] + K_a)$, $[FeOH^{2+}] = K_aC(Fe)/([H^+] + K_a)$ (1 point). After substitution into the equation for r we have the kinetic equation $r = (k_1[H^+] + k_2K_a)/([H^+] + K_a)C(Fe)\,C(Cl^-)$ and $r = k_{obs}C(Fe)\,C(Cl^-)$ (1.5 points). The observed rate constant, as it was obtained by Connick and Coppel, equals $k_{obs} = (k_1[H^+] + k_2K_a)/([H^+] + K_a)$.

For the values of k_{obs} at $[H^+] = 0.1$, 0.05 and 0.01 mol/L one can compose a system of three equations with three unknowns:

$709.2 = (0.1k_1 + k_2K_a)/(0.1 + K_a)$,

$1325.2 = (0.05k_1 + k_2K_a)/(0.05 + K_a)$,

$4458.0 = (0.01k_1 + k_2K_a)/(0.01 + K_a)$.

Solving the system of equations gives the value of $K_a = 6.8\cdot10^{-3}$ (1 point), $k_1 = 9.42$ (1 point), $k_2 = 11000$ (1 point). (5.5 point in total)

3. For the reaction $Fe^{3+} + HN_3 \rightleftarrows FeN_3^{2+} + H^+K = [FeN_3^{2+}][H^+]/([Fe^{3+}][HN_3])$. Since the reaction proceeds in two routes and is reversible, then its rate equals to the sum of the forward (r_+) and backward (r_-) reaction rates: $r = r_+ - r_-$. Considering both of these points, $r = k_3[Fe^{3+}][HN_3] + k_4[FeOH^{2+}][HN_3] - k_{-3}[H^+]C(FeN_3^{2+}) - k_{-4}C(FeN_3^{2+})$.

For the forward reaction $r_+ = (k_3[Fe^{3+}] + k_4[FeOH^{2+}])[HN_3]$. Since $[Fe^{3+}] = [H^+]\,C(Fe)/([H^+] + K_a)$ and $[FeOH^{2+}] = K_aC(Fe)/([H^+] + K_a)$, then substituting the concentrations in the expression for r_+ we obtain: $r_+ = (k_3[H^+] + k_4K_a)/([H^+] + K_a)\,C(Fe)[HN_3]$ (1 point).

In equilibrium $C(FeN_3^{2+}) = [FeN_3^{2+}]$, $r_+ = r_-$. Therefore, we get the equation $(k_{-3}[H^+] + k_{-4})[FeN_3^{2+}] = (k_3[H^+] + k_4K_a)/([H^+] + K_a)C(Fe)\,[HN_3]$ and $(k_{-3}[H^+] + k_{-4}) = C(Fe)\,[HN_3](k_3[H^+] + k_4K_a)/([H^+] + K_a)[FeN_3^{2+}]$. After substitution into the equation for the total rate, k_{-3}, k_{-4} disappear and we obtain $r = ((k_3[H^+] + k_4K_a)/([H^+] + K_a))C(Fe)[HN_3] - k_3[H^+] + k_4K_a)/([H^+] + K_a))(C(Fe)[HN_3]C(FeN_3^{2+})/[FeN_3^{2+}])$.

After conversion: $r = (k_3[H^+] + k_4K_a)/([H^+] + K_a)\,C(Fe)\,[HN_3]\,([FeN_3^{2+}] - C(FeN_3^{2+})/[FeN_3^{2+}])$ (1 point) or $r = k'_{obs}([FeN_3^{2+}] - C(FeN_3^{2+}))$. Since $C(Fe) = [Fe^{3+}]([H^+] + K_a)/[H^+]$, then $k'_{obs} = (k_3[H^+] + k_4K_a)/([H^+] + K_a)\,([H^+] + K_a)\,[Fe^{3+}]\,[HN_3]/[H^+][FeN_3^{2+}])$ and $k'_{obs} = (k_3[H^+] + k_4K_a)/K_a$ (0.5 points).

Using the table data, we have three equations:

$79.05 = (0.1k_3 + 6.8\cdot10^{-3}k_4)/0.59$,

$78.71 = (0.05k_3 + 6.8\cdot10^{-3}k_4)/0.59$,

$78.44 = (0.01k_3 + 6.8\cdot10^{-3}k_4)/0.59$.

Solving any pair of equations gives $k_3 = 4.0$ (0.5 points), $k_4 = 6800$ (0.5 points). Based on $K = k_3/k_{-3}$, $k_{-3} = k_3/K$ and $k_{-3} = 4.0/0.59 = 6.8$ (0.5 points).

(4 points in total)

4. In the case of anions of weak acids, the anation mechanism includes two stages: reversible $[Fe(H_2O)_6]^{3+} + N_3^-$ k'_{obs} $[Fe(H_2O)_5OH]^{2+} + HN_3$ (K_a), and irreversible $[Fe(H_2O)_5OH]^{2+} + HN_3 \rightarrow Fe(H_2O)_5N_3 + H_2O$ ($k_4 = 6800$) (0.5 points). Reaction equation with K_a can be obtained by the difference of two equations:

$[Fe(H_2O)_6]^{3+} \rightleftarrows [Fe(H_2O)_5OH]^{2+} + H^+$ (K_a),

$HN_3 \rightleftarrows H^+ + N_3^-$ (K_{a1}).

Then $K_a = K_a/K_{a1} = 6.8\cdot10^{-3}/1.9\cdot10^{-5} = 358$ and the reaction goes almost to completion (0.5 points).

The rate of the irreversible stage $r = k_4[Fe(H_2O)_5OH^{2+}][HN_3]$. From the equilibrium constant of the reversible stage $[Fe(H_2O)_5OH^{2+}] = K_a[Fe(H_2O)_6^{3+}][N_3^-]/[HN_3]$. Hence $r = k_4K_a[Fe(H_2O)_6^{3+}][N_3^-]$ (0.5 points) and the observed constant equals to $k''_{obs} = k_4K_a = 6800\cdot358 = 2.43\cdot10^6$ L·mol·s) (0.5 points). (2 points in total)

5. The activation energy is found from the Van't Hoff equation:

$$\ln(k_{T2}/k_{T1}) = (E_a/R)(1/T_1 - 1/T_2)E_a = (T_1T_2\cdot R\ln(k_{T2}/k_{T1}))/(T_2 - T_1)).$$

$E_a = (298\cdot313\cdot8.314\cdot\ln(275.9/83.6))/(313 - 298) = 61728$ J/mol. (1.5 points in total)

6. For a strong acid HCl:

1st stage – reversible $[Fe(H_2O)_6]^{3+} \rightleftarrows [Fe(H_2O)_5]^{3+} + H_2O$,

2nd stage – irreversible $[Fe(H_2O)_5]^{3+} + Cl^- \rightarrow Fe(H_2O)_5Cl^{2+}$ (0.5 points).

For a weak acid HN_3:

1st stage – reversible $[Fe(H_2O)_6]^{3+} + HN_3 \rightleftarrows Fe(H_2O)_6^{3+}, N_3^-$,

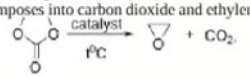
2nd stage – irreversible $[Fe(H_2O)_6]^{3+}N_3^- \rightarrow [Fe(H_2O)_5OH]^{2+} + HN_3$ (0.5 points).

The formation of an ion pair at the first stage, accompanied by the formation of a hydrogen bond, ensures the transfer of a proton and obtaining products at the second stage ($[Fe(H_2O)_5OH]^{2+}$ and HN_3), which further in the fast stage give the final products according to the reaction $[Fe(H_2O)_5OH]^{2+} + HN_3 \rightarrow [Fe(H_2O)_5N_3]^{2+} + H_2O$. (1 point in total)

Problem 3 (author Gulevich D.G.)

1. From the proposed structures, polymers **A** (polyamide 12) and **E** (copolymer of 4,4'-methylenediphenyldiisocyanate and THF) have less polar units than **B**, **C** and **D**, and therefore are more hydrophobic. (0.5 points per structure, 1 point in total)

2. **a)** Ethylene carbonate decomposes into carbon dioxide and ethylene oxide (**G**):



which is confirmed by data about M_{av}: $0.5M(\mathbf{G}) + 0.5M(CO_2) = 44$ g/mol, $M(\mathbf{G}) = 44$ g/mol. Therefore, **F** – polyethylene oxide $[-CH_2CH_2O-]_n$.

b) Let's write down the equations for the rates of each polymerization stage:

$V_i = k_i[H^+\mathbf{A}^-][\mathbf{G}]$, $V_p = k_p[M^+\mathbf{A}^-][\mathbf{G}]$, $V_a = k_a[M^+\mathbf{A}^-][\mathbf{G}]$, $V_t = k_t[M^+\mathbf{A}^-]$.

From the stationary approximation $V_i = V_t$ determine $[M^+\mathbf{A}^-] = k_i[H^+\mathbf{A}^-][\mathbf{G}]/k_t$. Then

$$V_p = \frac{k_ik_p[H^+\mathbf{A}^-][\mathbf{G}]^2}{k_t}$$

$$X_{av.} = \frac{V_p}{V_t + V_a} = \frac{k_p[\mathbf{G}]}{k_t + k_a[\mathbf{G}]} = \frac{0.94\cdot2.15}{7.5\cdot10^{-2} + 8.0\cdot10^{-3}\cdot2.15} = 22.$$

c)

(determination of **G** 1 point, 0.25 points for **F**, 1.75 points for obtaining the expression for the polymerization reaction rate, 0.5 points for calculating $X_{av.}$, 3.5 points in total)

3. **a)** According to the first law of thermodynamics $\Delta U = Q + A$. Calculate the amount of work done by the runner from the data on the generated power $-A = 235\cdot3600 = 846$ kJ. Negative values of internal energy, heat and work related to the fact that heat is released into the environment and work is done by the thermodynamic system itself. $Q = \Delta U - A = -3380 - (-846) = -2.534$ MJ. **b)** The air temperature in the room where the person is training will increase by

$$\Delta T = \frac{Q}{c_{air}\cdot m_{air}} = \frac{Q}{c_{air}\cdot\rho_{air}\cdot V_{air}} = \frac{2.534\cdot10^6}{1.005\cdot10^3\cdot1.165\cdot2500\cdot5} = 0.173\text{K}.$$

(calculation of the amount of heat 1.5 points, 1 point for ΔT, 2.5 points in total)

4. **a)** $V_{micro} = 4/3\pi r^3 = 4/3\cdot3.14\cdot(20\cdot10^{-6})^3 = 3.3\cdot10^{-14}$ m^3. Mass of one microsphere $m_{micro} = \rho_P\cdot V = 1200$ kg/m$^3\cdot3.3\cdot10^{-14}$ m$^3 = 3.96\cdot10^{-11}$ kg. The molar mass of polyethylene oxide is 22·44 g/mol = 968 g/mol, and its mass, taking into account the reaction yield, is equal to 2.15 mol·968 g/mol·0.95 = 1.98 kg. The total number of microspheres that are obtained and incorporate into the runner's clothing $N = 1.98/3.96\cdot10^{-11} = 5\cdot10^{10}$. **b)** Heat that can be absorbed by a material with a reversible phase change:

$$Q_{abs.} = \int_{T_{in}}^{T_{ph.ch.}} c_s dT + \Delta H_{p.ch.} + \int_{T_{p.ch.}}^{T_f} c_l dT.$$

Taking into account the number of the obtained microspheres $Q_{abs} = 8.48\cdot10^{-11} \cdot 5\cdot10^{10}\cdot(311 - 307) + 6.11\cdot10^{-9}\cdot5\cdot10^{10} + 6.28\cdot10^{-11}\cdot5\cdot10^{10}\cdot(313 - 311) = 329$ kJ:

$$\frac{329 \cdot 0.8}{2534} \cdot 100\% = 10\%.$$

(calculation of the polymer mass 0.5 point, 1.5 point for the number of microspheres, 1.75 point for the amount of absorbed heat, 3.75 points in total)

5. **a)** Calculate the generated output voltage:

$$U = \frac{\gamma Ph}{\varepsilon\varepsilon_0} = \frac{\gamma mgh}{\varepsilon\varepsilon_0 S} = \frac{49.6 \cdot 10^{-12} \cdot 55 \cdot 9.8 \cdot 30 \cdot 10^{-6}}{10 \cdot 8.85 \cdot 10^{-12} \cdot 2 \cdot 0.0189} = 0.24\ \text{V}$$

The current strength is determined by power data: $I = W/U = 10^{-3}/0.24 = 4.2$ mA.

$$\frac{4.2 \cdot 10^{-3}}{130 \cdot 10^{-3}} \cdot 100\% = 3.2\%$$

b) For an hour of running, a person can charge the heart rate sensor for .

c) Determine **L** – $C_xH_yX_z$. Based on mass spectrometry data, it can be assumed that the molar mass **L** is equal to 64 g/mol, then $x = 0.375 \cdot 64/12 = 2$, $y = 0.031 \cdot 64 \approx 2$. Further, $zA_r(X) = 64 - 26 = 38$ g/mol, consequently, X – F, $z = 2$ and **L** – $C_2H_2F_2$. Mass spectrometry data will help to establish the isomer of difluoroethylene. Let us determine which ions correspond to the peaks in the spectrum:

m/z	ion
64	$\mathbf{C_2H_2F_2^+}$
63	$C_2H_2F_2^+ \rightarrow \mathbf{C_2HF_2^+} + H$
50	$C_2H_2F_2^+ \rightarrow \mathbf{CF_2^+} + CH_2$
14	$C_2H_2F_2^+ \rightarrow \mathbf{CH_2^+} + CF_2$

Based on the presence of CF_2^+ and CH_2^+ ions, it can be concluded that **L** is 1,1-difluoroethylene. Therefore, **K** is poly(1,1-difluoroethylene).

(calculation of the output voltage - 1.25 points, 0.75 points for the current strength, 0.25 points for the charge percentage, 1 point for determination **K** and **L**, 1 point for determination the isomer of **L**, 4.25 points in total).

SECTION II. ORGANIC CHEMISTRY

Problem 1 (author Shved A.M.)

1. Since "carbon oxide" **X1** is formed by the reaction of ozone (O_3) with a simple substance **A**, it can be concluded that **A** is one of the allotropic modifications of carbon, and the formula of **X1** is C_xO_y.

$$w(O)_{C_xO_y} = \frac{16.00y}{12.01x + 16.00y} = 0.0624 \Rightarrow \frac{x}{y} = 20$$

Having a mass fraction of oxygen in **X1**, we get:

When **X1** decomposes, an oxygen molecule (O_2) is released, and a new "carbon oxide" **X2** is formed. So, **X1** most likely contains at least 3 oxygen atoms, and if $y = 3$, then $x = 60$. This correlates with the use of ozone to oxidize fullerene C_{60}. Then **A** – C_{60}, **X1** – $C_{60}O_3$, and **X2** – $C_{60}O$ (0.5 points per formulas of **A**, **X1**, and **X2**, 0.25 points for calculations, 1.75 points in total).

2. The simplest dicarboxylic acid **B** is ethanedioic (oxalic) acid. When phosphorus (V) chloride and silver carbonate react with it, diacyl chloride **C** and silver oxalate (**D**) are formed, respectively. When **C** and **D** react with each other, it is logical to assume a 1:1 reaction with the release of 2 equivalents of AgCl and the closure of the cycle. This gives the desired structure of the "carbon oxide" **X4** with the molecular formula C_4O_6.

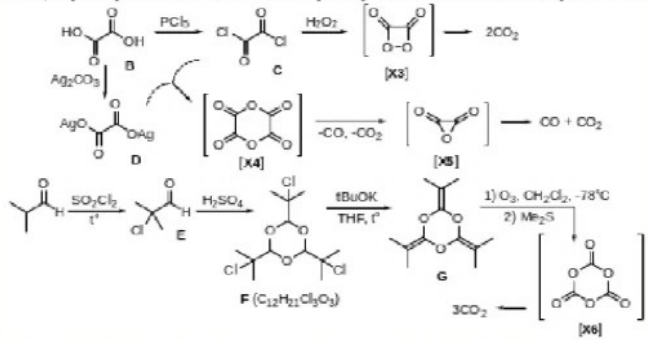
This substance successively decomposes to form an equimolar mixture of two gases and substance **X5**, which then also decomposes into the same mixture of gases. Hence, **X4** is formally an **X5** dimer with the formula C_2O_3. The same formula is the sum of the formulas of the gases CO and CO_2, the actual decomposition products. Based on the structure of **X4**, the most likely structure of **X5** is cyclic oxalic acid anhydride.

The stage of **X3** formation from **C** and H_2O_2 can be considered analogous to the reaction of hydrolysis of acyl chlorides. Allowing for an equimolar ratio of reactants and subtracting 2 molecules of HCl as a product, we get **X3** – C_2O_4 with a cyclic structure having a peroxide bridge. Then the **X3** decomposition product, the gas with a density, and therefore a molar mass, higher than the air, is carbon dioxide CO_2.

Thus, formally, **X3** is a CO_2 dimer. **X6** also decomposes with the formation of the same gas, and then its molecular formula is C_nO_{2n}.

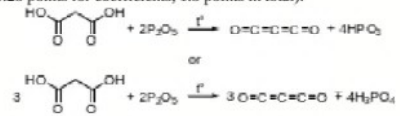
The stage of formation of **E** from isobutyraldehyde is most likely the chlorination in the alpha position to the aldehyde group, i.e., **E** – α-chloroisobutyraldehyde (C_4H_7ClO). A threefold increase in the number of all atoms during the synthesis of **F** corresponds to trimerization to form the structure of trioxane **F**. The last stage of synthesis reminds the ozonolysis of double bonds in **G**, which can be formed by elimination of **F** with potassium *tert*-butoxide. Overall, **X6** has the formula C_3O_6 and is a carbon dioxide trimer.

Schemes (0.5 points per the structure of **B**–**G**, 0.75 points per the structure of **X3**–**X6**, 6 points in total):



3. The adjacent homolog of oxalic (ethanedioic) acid is malonic (propanedioic) acid $C_3H_4O_4$. Phosphorus (V) oxide serves as a dehydrating agent so that when two water molecules are eliminated from $C_3H_4O_4$, a malonic anhydride C_3O_2 is formed.

Reaction equation (0.25 points per formula of $C_3H_4O_4$, C_3O_2, HPO_3, or H_3PO_4, 0.25 points per structures of $C_3H_4O_4$ and C_3O_2, 0.25 points for coefficients, 1.5 points in total):

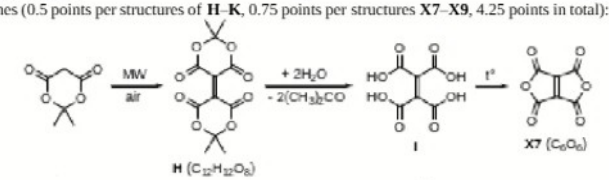


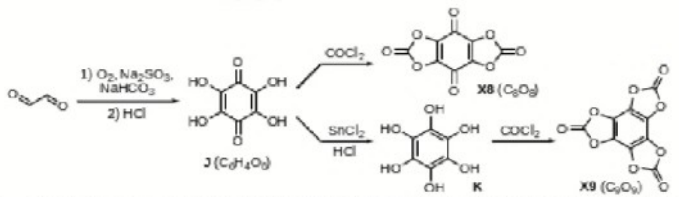
4. The lightest carbon oxide is carbon monoxide CO. Thus, **X7**–**X9** have molecular formulas with equal numbers of carbon and oxygen atoms.

The molecular formula of the starting material in the synthesis of **H** is $C_6H_8O_4$. Since the reaction takes place in the air, and **H** contains twice the number of carbon and oxygen atoms but a smaller number of hydrogen atoms, it can be assumed that this is an oxidative dimerization reaction. **H** has only 1 type of protons – probably methyl groups protons, and 4 types of carbon atoms – just like in the initial compound. This allows for determining the symmetric structure of compound **H**. The next step is the hydrolysis of the ester bonds, resulting in tetracarboxylic acid **I** ($C_6H_4O_8$). It contains 1 type of hydrogen atoms and only 2 types of carbon atoms, which correlates with the number of signals in NMR spectra. Dehydration of **I** with the elimination of 2 water molecules and the closure of 2 cycles gives the desired "carbon oxide" **X7**, which has the formula C_6O_6 and is an anhydride of acid **I**.

The formation step of **K** from **J** is a reduction reaction. Given the molecular formula of **J** and the fact that **K** has only 1 type of carbon and hydrogen atoms in its structure, it can be assumed that the reduction of **J** led to symmetrical hexahydroxybenzene $C_6H_6O_6$. **J** could then have the structure of a tetrahydroxy-substituted *p*-benzoquinone, which has 1 type of H atoms and 2 types of C atoms. Hence, **X8** and **X9** are carbonic acid and polyphenols esters with the formulas C_8O_8 (tricyclic compound) and C_9O_9 (tetracyclic compound), respectively.

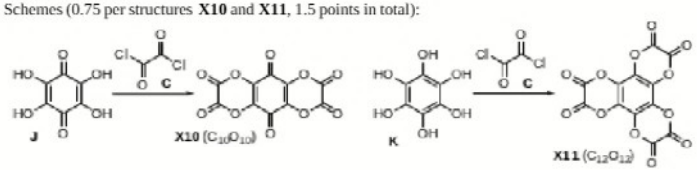
Schemes (0.5 points per structures of **H**–**K**, 0.75 points per structures **X7**–**X9**, 4.25 points in total):





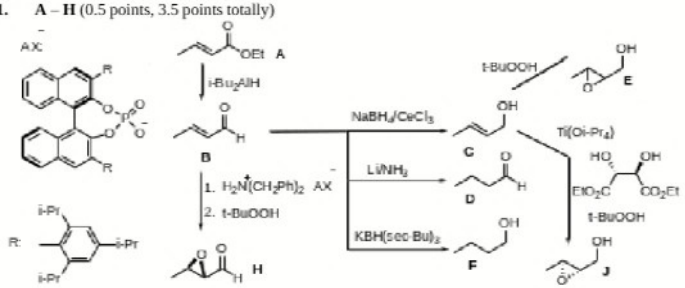
5. Knowing the number of carbon atoms in **X10** and **X11** and the fact that their empirical formula also corresponds to the CO formula, it is possible to unambiguously determine the molecular formulas: **X10** – $C_{10}O_{10}$ and **X11** – $C_{12}O_{12}$. It is not difficult to draw the structures if the formulas of **C**, **J**, and **K** have been determined. Or conversely, this question can help determine the formulas of **K** and **C** if they have not been found earlier.

Schemes (0.75 per structures **X10** and **X11**, 1.5 points in total):



Problem 2 (author Shved E.N.)

1. **A** – **H** (0.5 points, 3.5 points totally)



2. Structures of **H** and **G** are given above (0.5 points each, 1 point totally)

3. Chemoselective reactions: **A** to **B**, **B** to **C**, **B** to **D**, **C** to **E**, **C** to **J**, **B** to **H**. Regioselective reactions: none. Stereoselective reactions: transformation of **C** into **J**, **B** into **H**. Enantioselective reactions: transformation of **C** into **J**, **B** into **H** (0.25 points per reaction, total: 2.5 points, for wrong answers - 0.25 point penalty).

4. At the first stage, the acetal of **M** is hydrolyzed in the presence of acid catalyst. The resulting aldehyde reacts with the diene containing electron-donating ethoxy and silyloxy groups. A chiral Lewis acid is formed from chiral binaphthol and titanium isopropylate to provide enantioselective formation of the stereocenter at the hydroxyl group. The subsequent reaction with tetrabutylammonium fluoride makes it possible to remove the silyl protection. After the conversion of enol to the carbonyl form **N** was isolated. Then, the enantioselective reduction of the carbonyl group is carried out in the presence of Et_2BOMe. This Lewis acid provides the formation of a boron chelate, through the formation of which **III** is obtained. Then, the silyl protection of the alcohol groups of t-BuMe $_2$SiCl is carried out and then the ester group is selectively reduced to aldehyde **P**. The next step is enantioselective cyclization in the presence of an organocatalyst which is phosphoric acid based on substituted binaphthol and proline. After reduction of the aldehyde fragment compound **IV** is formed. It is converted to **Q** by enzymatic acylation of the sterically more accessible alcohol group in the presence of lipase. The remaining unprotected alcohol fragment is oxidized using the Dess-Martin reagent to an aldehyde, and then the Wadsword-Horner-Emmons reaction is carried out in the presence of sodium hexamethyldisilazide as a base to give compound **S**. The ketone fragment in its structure is enantioselectively reduced using a chiral oxazaborolidine (reduction CBS). **T** is obtained after silylation of the resulting alcohol. Hydrolysis of the acetate in the presence of potash and subsequent oxidation with a Dess-Martin reagent afford the aldehyde. Under Wittig reaction conditions with a phosphorus ylide prepared from a phosphonium salt in the presence of a base, the lower chain of the target structure is constructed to form **W**. Note that the phosphonium salt used for this has the desired configuration of two double bonds. The prepared double bond has Z-configuration, which is typical for non-activated phosphorus ylides. The synthesis of X is completed by the removal of silyl groups and hydrolysis of the ester. (1 point for each structure, total 7 points)